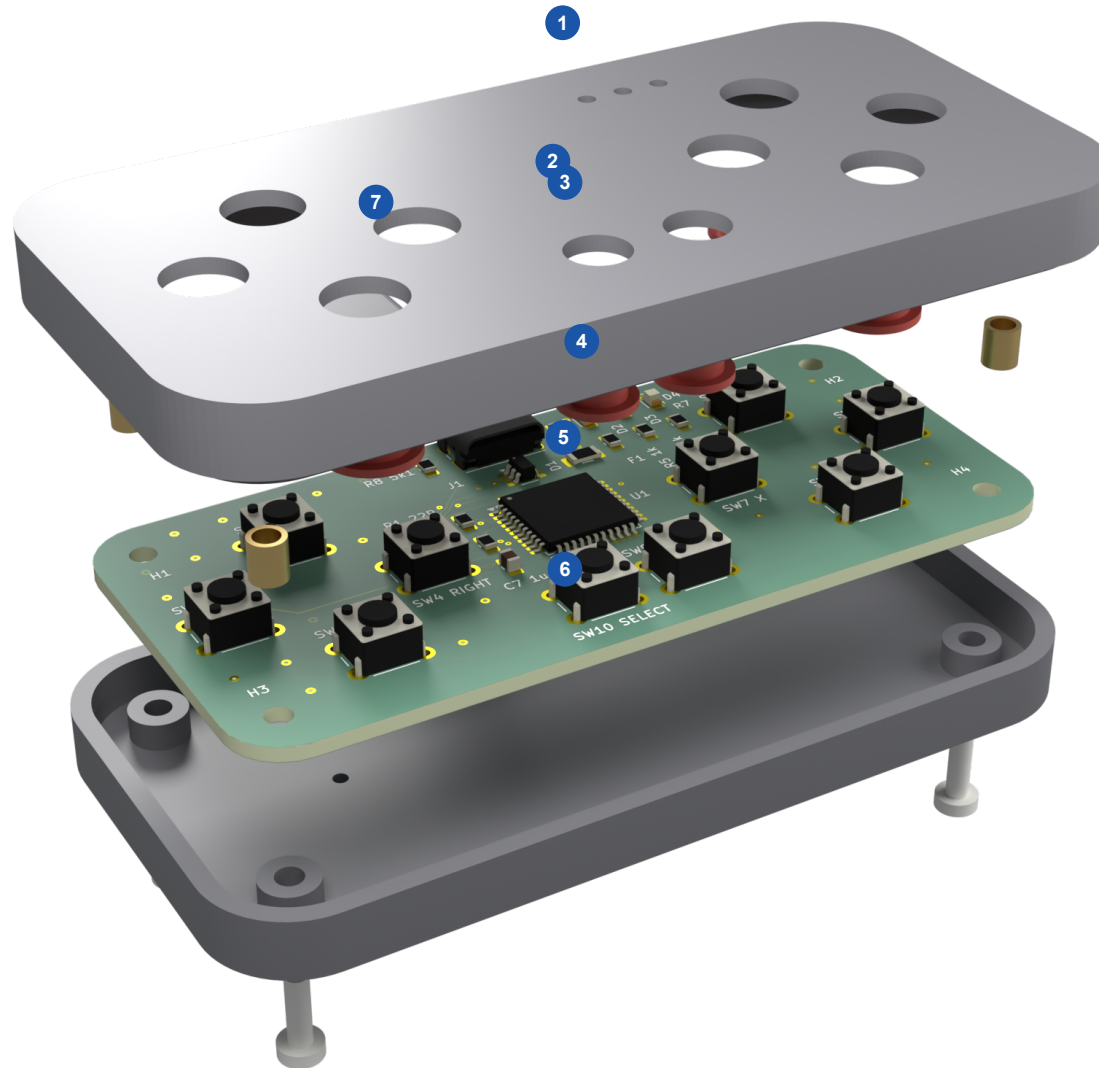


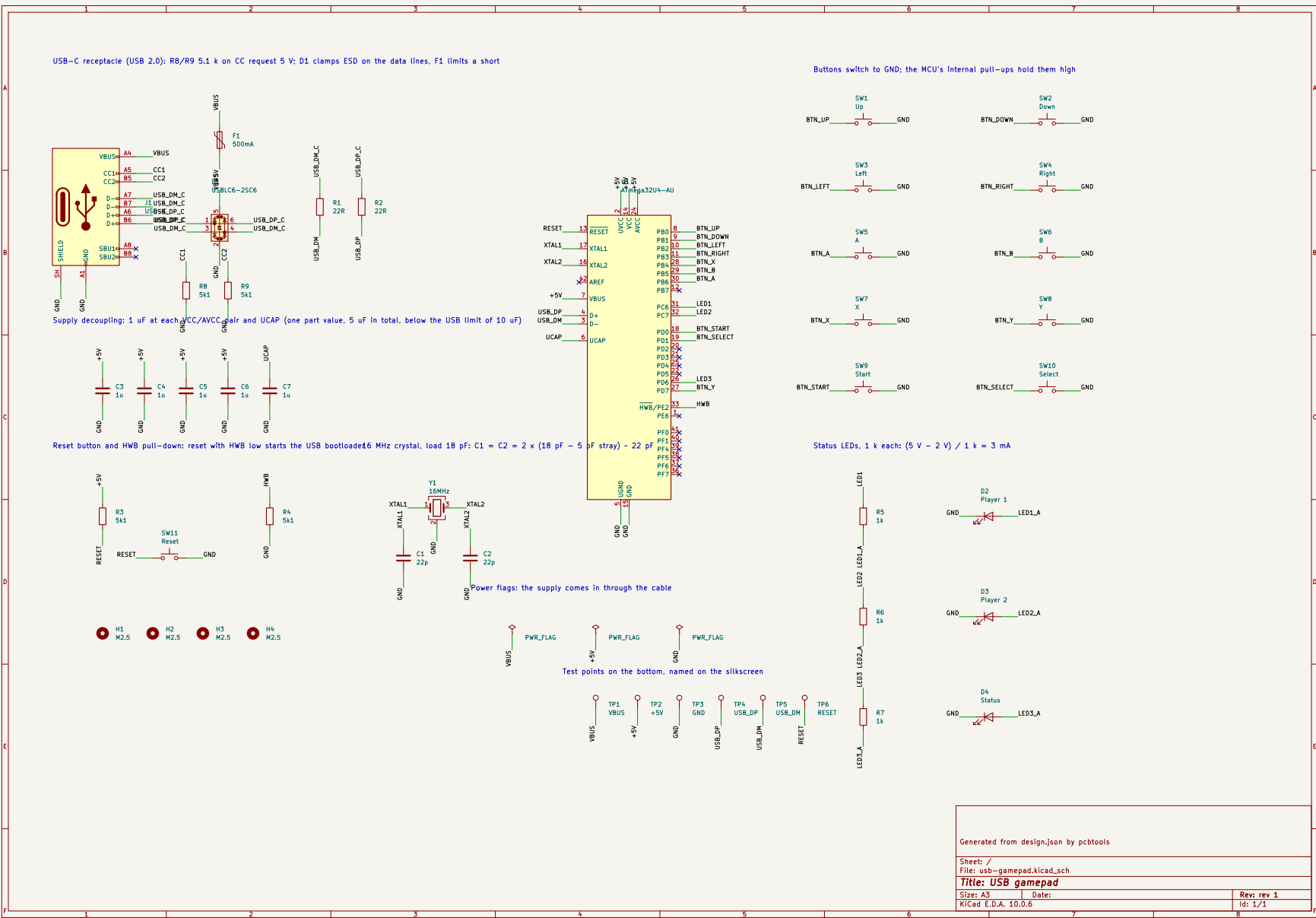
USB gamepad – overview

Purpose: a small USB gamepad for PC games: D-pad, A/B/X/Y, START, SELECT, three status LEDs, USB-C to the PC, recognised without a driver (HID). **Series:** 25 devices, hand soldered. **Budget:** 450 € for the series (fictional). **Cost:** ≈ 301 € for 25 devices (12.05 € each): parts 224 €, boards 62 €, PETG 16 €.



1 [top shell, PETG](#) · 2 [button caps, PETG](#) · 3 [heat-set inserts M2.5](#) · 4 [board, 86 × 40 mm](#) · 5 [base shell, PETG](#) · 6 [screws M2.5 × 8](#) · 7 [USB-C cable \(symbolic\)](#). Exploded in assembly order: base, board, inserts and caps, top; screws from below.

Schematic

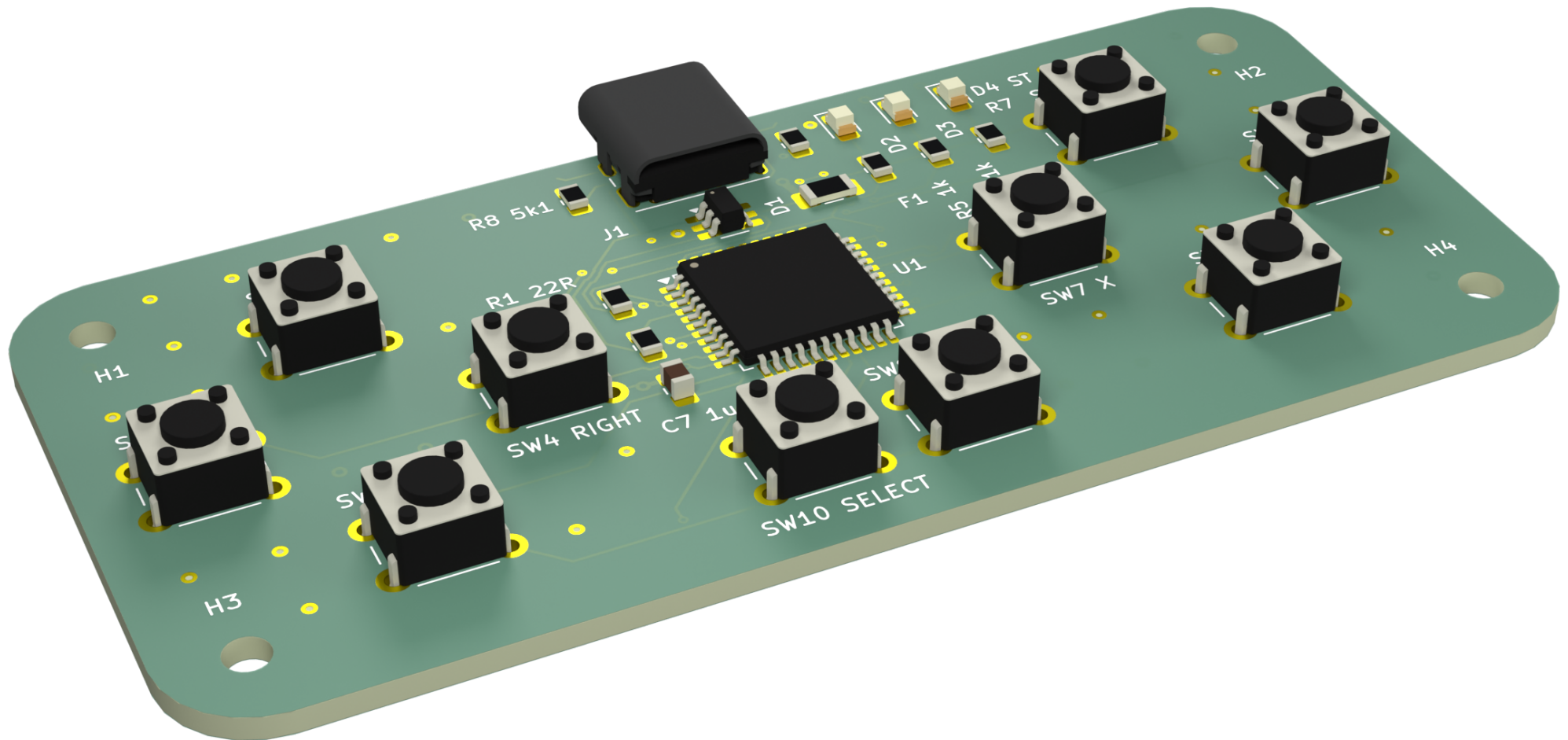


Operating points per part

Worst case VBUS 5.25 V. Example values from typical datasheets; for a real build each one comes from the datasheet of the bought part (CI-1). Total dissipation ≈ 0.13 W: no thermal limit inside the case (rise < 1 K).

Ref	Function	Case	Load	Limit	Use	Verdict
U1	MCU supply	VBUS 5.25 V max.	5.25 V	operating 2.7–5.5 V	95 %	within CI-2; inside the operating range
U1	MCU current	16 MHz, USB active	≈ 15 mA	USB budget 50 mA declared	30 %	within CI-2
D2–D4	status LEDs	on	3 mA each	20 mA	15 %	within CI-2
R5–R7	LED resistors 1 k	LED on	9 mW	125 mW	7 %	within CI-2
R3	reset pull-up 5.1 k	button pressed	4.9 mW	125 mW	4 %	within CI-2
R8, R9	CC pull-downs 5.1 k	host 5 V on CC	≈ 0.2 mW	125 mW	1 %	within CI-2; value set by USB-C
F1	PTC 500 mA	normal operation	≈ 25 mA	hold 500 mA	5 %	within CI-2; trips at a short
C3–C7	1 μ F X7R 25 V	supply	5.25 V	25 V	21 %	within CI-2; 5 μ F in total, USB limit 10 μ F
C1, C2	22 pF C0G 50 V	crystal	≈ 5 V peak	50 V	10 %	within CI-2; $2 \times (18 - 5)$ pF = 26 pF $\rightarrow$ 22 pF
J1	USB-C receptacle	device current	≈ 25 mA	≥ 1.5 A per pin	2 %	within CI-2
SW1–SW 11	tactile switches	pull-up current	0.14 mA	50 mA	1 %	within CI-2

Board, assembled

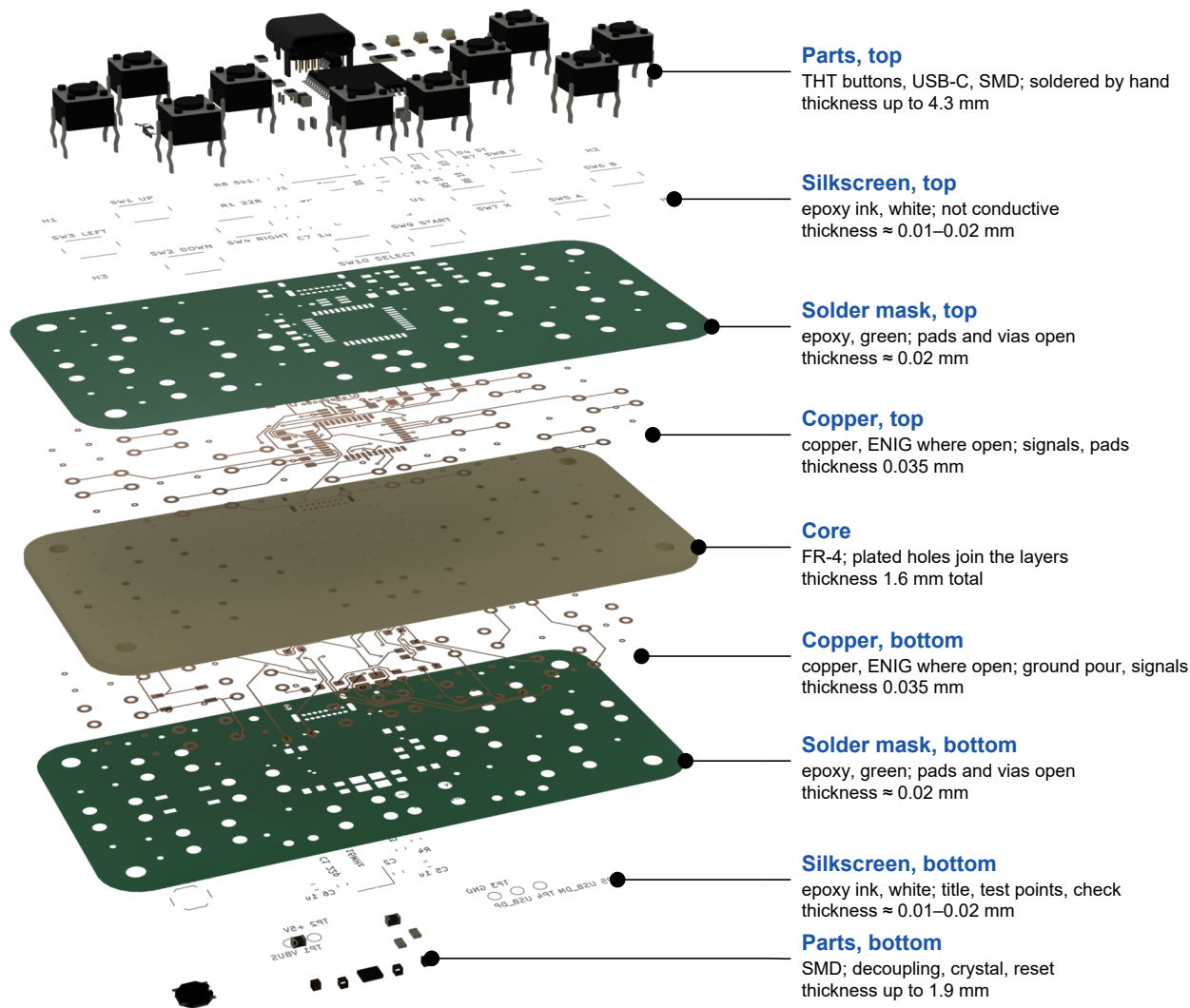


Rendered from the KiCad 3D export: 86 × 40 × 1.6 mm, 2 layers, ENIG. Hand-solder footprints, THT buttons and USB-C, decoupling, crystal and reset button on the bottom.

Layer stack

86.0 × 40.0 × 1.6 mm

34.1 cm², bare ≈ 10 g (FR-4 1.85 g/cm³)

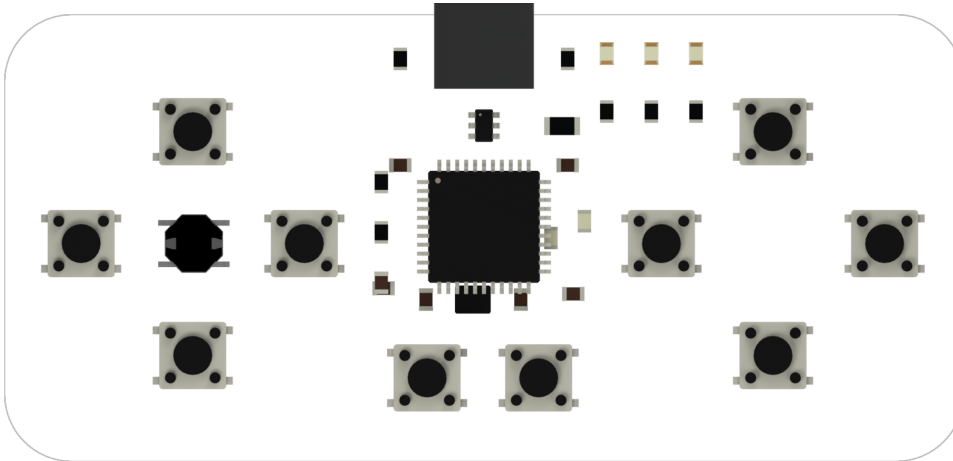


Exploded view from above; every layer to scale, only the gaps are invented. The bottom side is shown as it lies in the product (seen from above). Each layer and label links to its detail.

Layers in detail, top to bottom

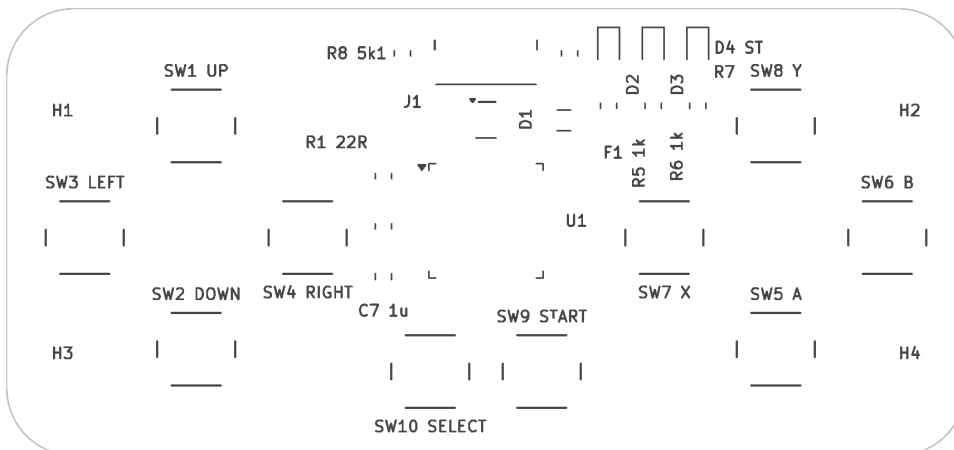
Parts, top

Buttons, USB-C receptacle, MCU, USB resistors and ESD, LEDs. Plan view of the 3D models at the scale of the layers below.



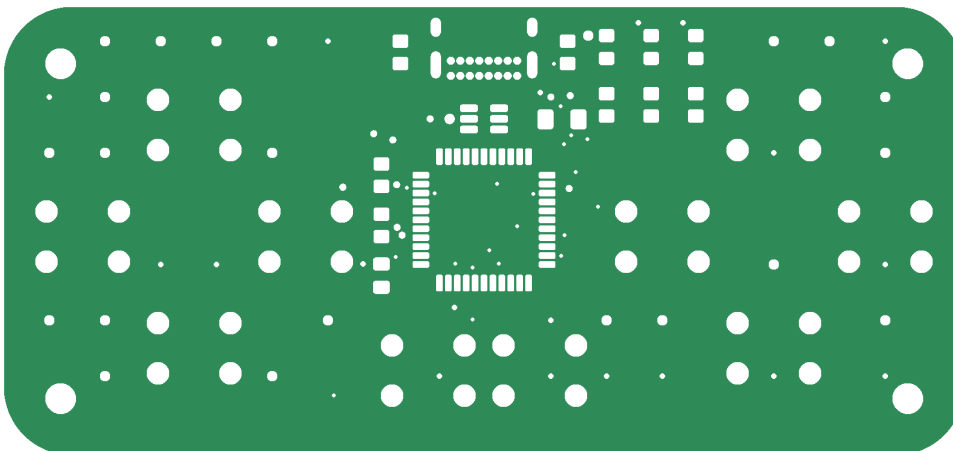
Silkscreen, top (F.Silks)

Every reference with its value or function next to its part; button names under the buttons.



Solder mask, top (F.Mask)

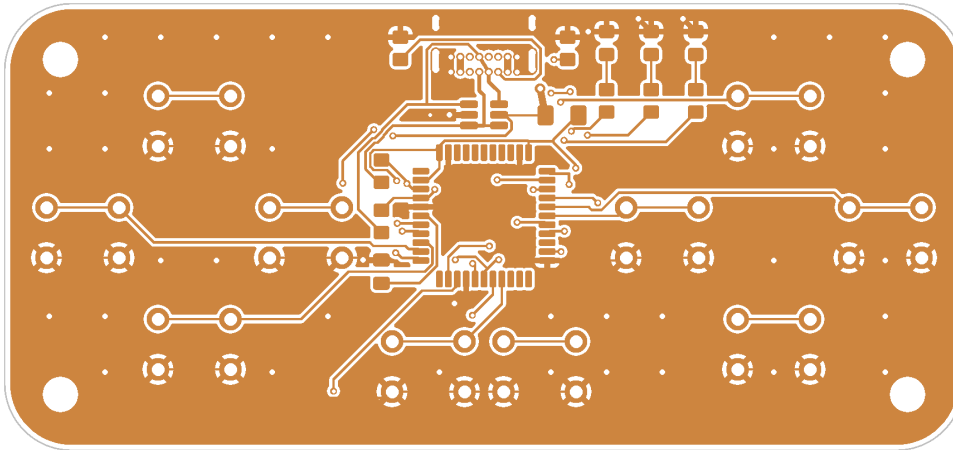
Green = mask. Open at every pad and at the vias not under a part.



Layers in detail, top to bottom

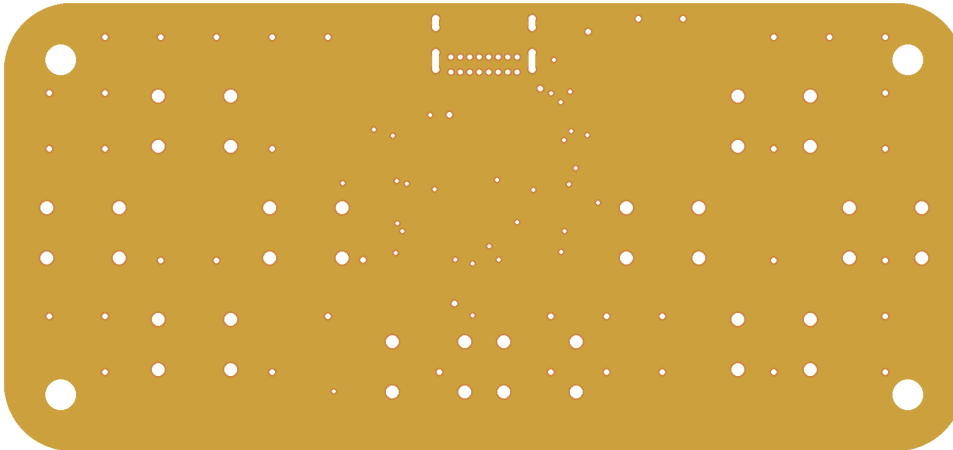
Copper, top (F.Cu)

Signals 0.3 mm, supply 0.6 mm, clearance 0.2 mm (0.15 mm only between the USB-C pins).



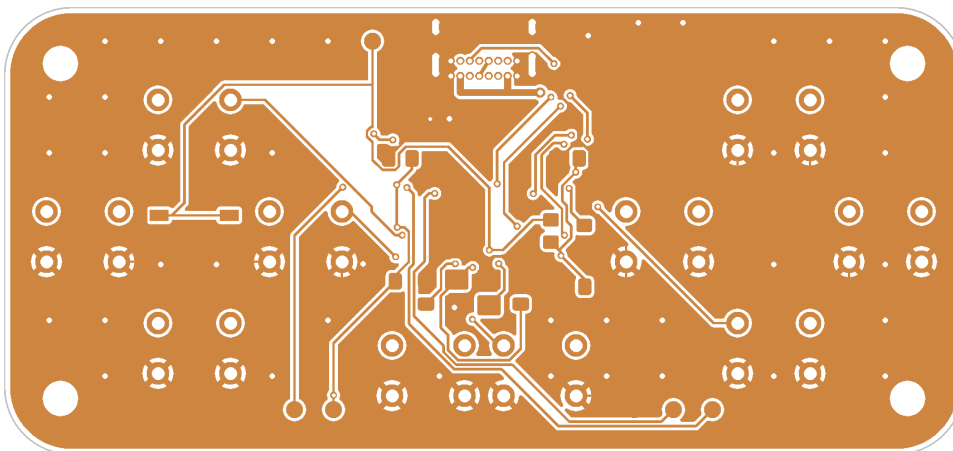
Core with holes

FR-4 1.6 mm; THT buttons, USB-C pins and shield, vias, four M2.5 holes.



Copper, bottom (B.Cu)

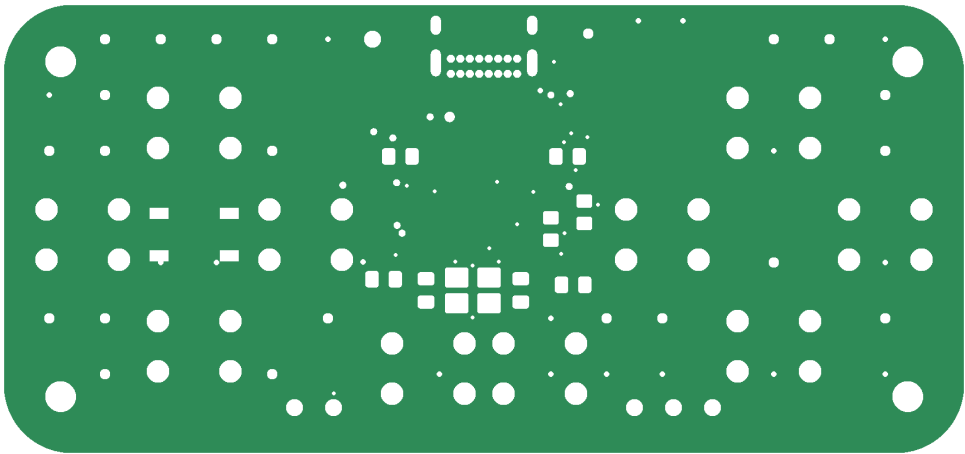
Ground pour with the remaining signals; seen from above.



Layers in detail, top to bottom

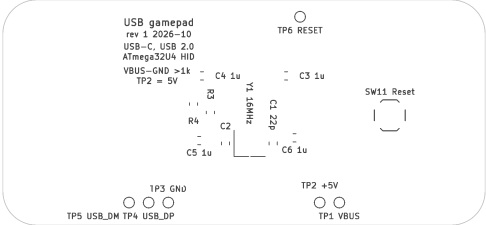
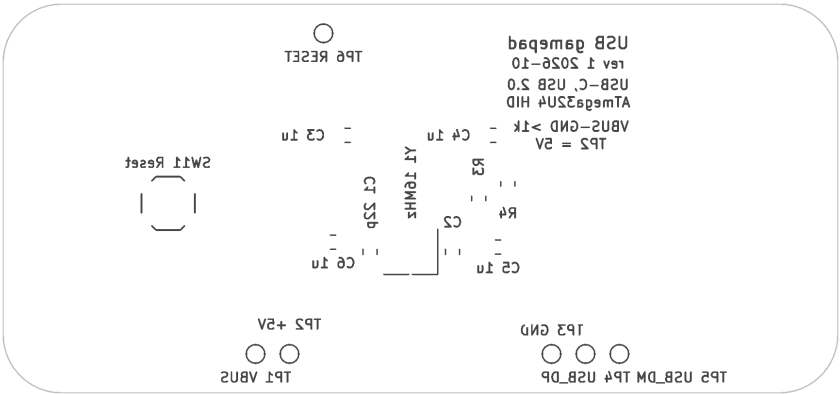
Solder mask, bottom (B.Mask)

Open at the pads, test points and open vias.



Silkscreen, bottom (B.Silks)

Title, revision, the first check after soldering, test point names, references of the bottom parts. Left as stacked, right readable.



readable, board turned over

Parts list: Distributor A

Distributor A (fictional): 8 of 13 distinct parts, shipping 4.95 € net.

Ref	Part	Data / function	Order no.	Series need / stock	Delivery	Unit price, net
U1	ATmega32U4-AU (TQFP-44)		EX-85CDC	25 → 25 / 10 000+	2 working days	4.400
Y1	Crystal 16 MHz 3225 18 pF		EX-39E73	25 → 25 / 10 000+	2 working days	0.350
D2 D3 D4	LED 0805 green		EX-BE2AA	75 → 75 / 10 000+	2 working days	0.060
F1	PTC fuse 500 mA 1206		EX-382B5	25 → 25 / 10 000+	2 working days	0.140
SW1 SW2 SW3 SW4 SW5 SW6 SW7 SW8 SW9 SW10	Tactile switch 6x6 THT		EX-C7C72	250 → 250 / 10 000+	2 working days	0.060
SW11	Tactile switch SMD 4x4 (TL3342 type)		EX-B20D8	25 → 25 / 10 000+	2 working days	0.300
J1	USB-C receptacle USB 2.0 THT		EX-5E7CD	25 → 25 / 10 000+	2 working days	0.700
D1	USBLC6-2SC6 (SOT-23-6)		EX-5E98C	25 → 25 / 10 000+	2 working days	0.300

Example data: supplier, links, order numbers and prices are invented.

Parts list: Distributor B

Distributor B (fictional): 5 of 13 distinct parts, shipping 6.50 € net.

Ref	Part	Data / function	Order no.	Series need / stock	Delivery	Unit price, net
C3 C4 C5 C6 C7 C8	C0805 1u X7R 25V		EX-D5223	125 → 200 / 10 000+	2 working days	0.006
C1 C2	C0805 22p C0G 50V		EX-E5C6F	50 → 100 / 10 000+	2 working days	0.004
R5 R6 R7	R0805 1k 1%		EX-A6087	75 → 100 / 10 000+	2 working days	0.003
R1 R2	R0805 22R 1%		EX-7ADFC	50 → 100 / 10 000+	2 working days	0.003
R3 R4 R8 R9	R0805 5k1 1%		EX-F6101	100 → 100 / 10 000+	2 working days	0.003

Example data: supplier, links, order numbers and prices are invented.

Parts list: special parts, PCB, enclosure material

Mechanics (fictional prices): 4 × M2.5 × 8 screws, 4 × heat-set inserts M2.5 × 4, PETG ≈ 31 g per device at full infill (≈ 15 g with 3 walls and 20 % infill).

PCB manufacture and assembly: offers

Board 86 × 40 mm, 2 layers, FR-4 1.6 mm, 35 µm, ENIG, green mask, white silkscreen, 25 pieces. Example offers, names and prices fictional; landed totals include shipping and import costs to the delivery country.

PCB

Fab	Settings	Qty	Landed total	Lead time	Notes
Fab 1 (EU)	2 L, 1.6 mm, ENIG, green/white	25	61.90 €	10 working days	cheapest; shipping included
Fab 2 (overseas)	2 L, 1.6 mm, ENIG	25	68.40 €	4 days + 8 days shipping	ENIG surcharge, import VAT
Fab 3 (local express)	2 L, 1.6 mm, ENIG	25	182.00 €	3 working days	only worth it when time matters

Assembly

Assembler	Basis	Qty	Total	Lead time	Notes
Assembler 1 (overseas, turnkey)	both sides, THT by hand at the assembler	25	≈ 240 €	3 weeks	parts sourced by the assembler; two-sided adds a set-up fee
Assembler 2 (EU)	consigned parts	25	≈ 410 €	2 weeks	parts from the distributor list

Design to cost

Every line of the BOM lists the parts that would meet the requirement; pcbtools cost prices each combination for the series and keeps the cheapest. Per-part fee 2.00 € for each distinct part (reel, feeder, sorting), placement 0.01 €. The board was then laid out with the optimised parts.

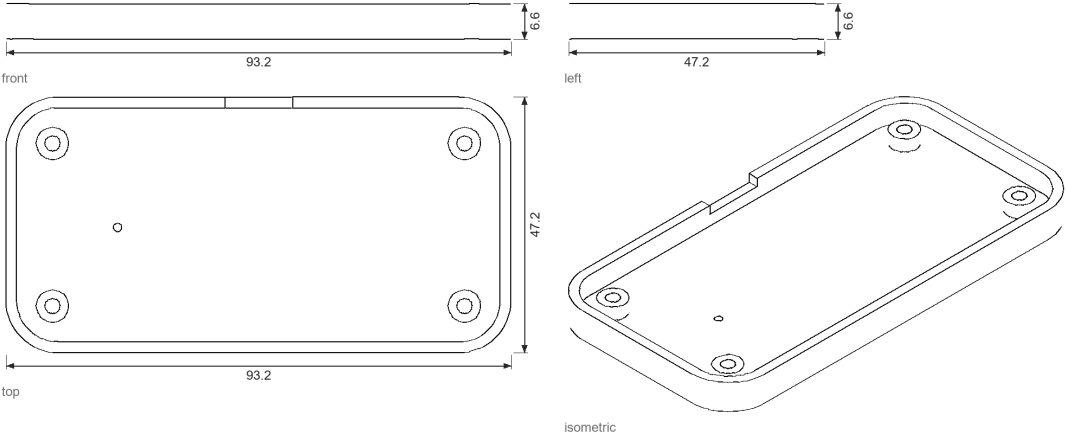
Version	Parts	Shipping	Per-part fees	Total	Distinct parts	Area
first version	180.50	11.45	41.00	232.95	16	64.5 mm²
optimised, built	176.75	11.45	34.75	222.95	13	54.9 mm²

Refs	Function	First version	Optimised	Why it still meets the requirement
C3, C4, C5, C6, C7, C8	supply decoupling, UCAP, bulk	4 x 100 nF + 1 uF (UCAP) + 10 uF bulk	5 x 1 uF, no bulk capacitor	1 uF at each supply pin decouples as well as 100 nF for a 16 MHz MCU; 5 uF in total stays below the 10 uF a USB device may present; UCAP needs 1 uF anyway
R3, R4	reset pull-up, HWB pull-down	R0805 10k 1%	R0805 5k1 1%	same value as the CC resistors; reset pull-up and HWB pull-down only need to be well below the 50 k internal resistors

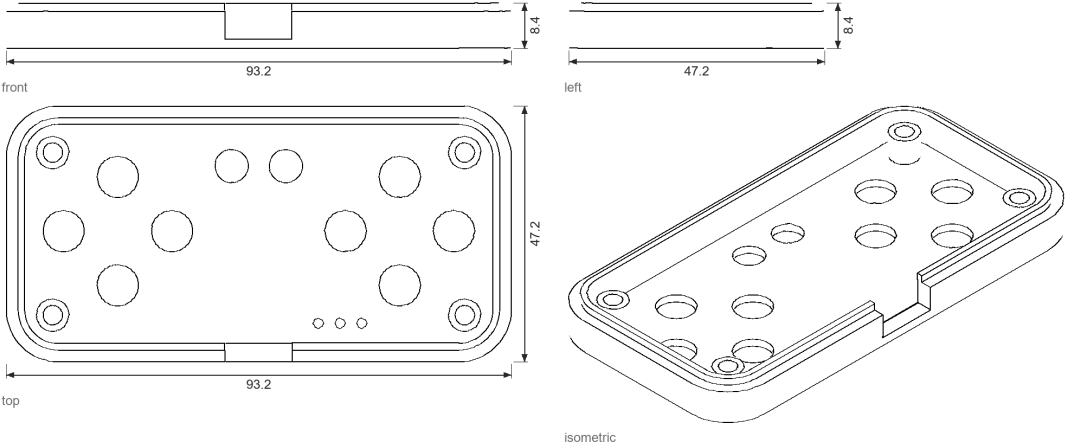
Enclosure, technical drawing

USB gamepad enclosure, PETG
First-angle projection (ISO). Dimensions in mm. Visible edges only.

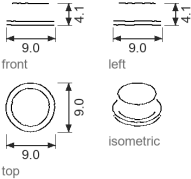
Base



Top



Cap



Enclosure: fasteners, material, heat, radio

Size: $93.2 \times 47.2 \times 13.5$ mm; base 2 mm floor, 2 mm walls; the top's lip (1.2 mm) locates it inside the base walls. Board clamped between the base bosses and the top bosses.

Screws and inserts: $4 \times M2.5 \times 8$ from below into heat-set inserts $M2.5 \times 4$ in the top bosses; 3 mm of thread in the insert. Load: a dropped gamepad (≈ 60 g, 1 m onto a hard floor) stays far below the pull-out force of M2.5 inserts in PETG; hand-tight only.

Caps: ten printed caps with a retaining flange under the top plate; the plunger of each tactile switch carries its cap, 1.5 mm proud of the surface.

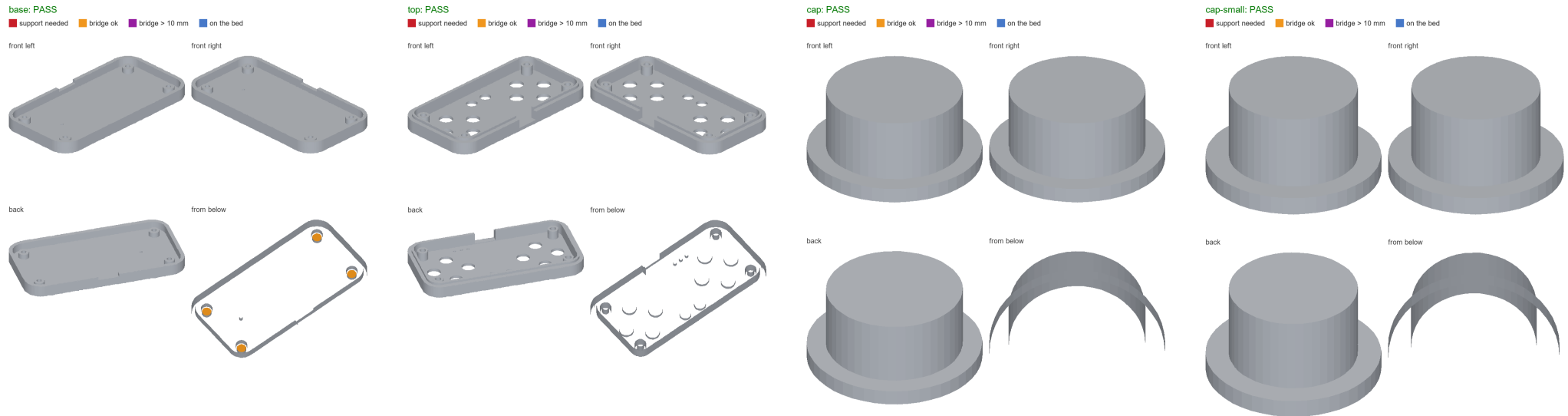
Material: PETG (impact resistant, 70 °C service temperature), ≈ 15 g per device with 20 % infill; ≈ 0.62 € at 25 €/kg (fictional).

Heat: ≈ 0.13 W in a $93 \times 47 \times 14$ mm case: temperature rise below 1 K; no vents needed.

Radio: none on the board. USB-C shield to GND.

Openings: USB-C 12.4×7 mm for the cable's overmould, three $\varnothing 1.8$ mm LED windows, a $\varnothing 1.6$ mm hole in the base for the reset button (paper clip).

Enclosure: printability (FDM)



base: PASS – overhang beyond 45°: 0.0 mm²; longest bridge 4.99 mm (limit 10); islands 0; layers with features below 0.8 mm: 0; first layer 4239.2 of 4317.5 mm².

top: PASS – overhang beyond 45°: 0.0 mm²; longest bridge 0 mm (limit 10); islands 0; layers with features below 0.8 mm: 0; first layer 3891.8 of 3891.8 mm².

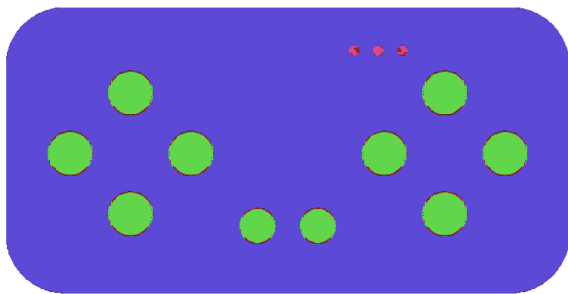
cap: PASS – overhang beyond 45°: 0.0 mm²; longest bridge 0 mm (limit 10); islands 0; layers with features below 0.8 mm: 0; first layer 63.4 of 63.4 mm².

cap-small: PASS – overhang beyond 45°: 0.0 mm²; longest bridge 0 mm (limit 10); islands 0; layers with features below 0.8 mm: 0; first layer 44.1 of 44.1 mm².

Base printed floor down, top printed plate down, caps flange down; 0.2 mm layers, 0.4 mm nozzle. The screw seats print as a bridge over a 0.2 mm membrane.

Enclosure: virtual assembly

top: red 0.8 % of visible area



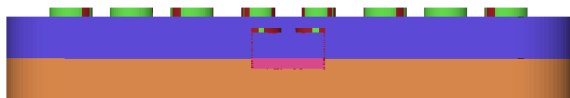
bottom: red 0.1 % of visible area



front: red 1.1 % of visible area



back: red 1.7 % of visible area



left: red 0.6 % of visible area



right: red 0.6 % of visible area



Intersections: 21 pairs checked, 21 pass.

Pair	Volume	Result
base / top	0.0 mm ³	pass
base / caps	0.0 mm ³	pass
base / board	0.0 mm ³	pass
base / parts	0.0 mm ³	pass
base / screws	3.609 mm ³	allowed
base / inserts	0.0 mm ³	pass
top / caps	0.0 mm ³	pass
top / board	-0.0 mm ³	pass
top / parts	0.0 mm ³	pass
top / screws	0.0 mm ³	pass
top / inserts	0.0 mm ³	allowed
caps / board	0.0 mm ³	pass
caps / parts	0.0 mm ³	pass
caps / screws	0.0 mm ³	pass
caps / inserts	0.0 mm ³	pass
board / parts	0.0 mm ³	pass
board / screws	0.0 mm ³	pass
board / inserts	0.0 mm ³	pass
parts / screws	0.0 mm ³	pass
parts / inserts	0.0 mm ³	pass
screws / inserts	0.0 mm ³	allowed

Closure views: inward-facing surfaces in red; visible red per view: top 0.8 %, bottom 0.1 %, front 1.1 %, back 1.7 %, left 0.6 %, right 0.6 %.

Order: 1. solder the board (bottom side first), 2. press the inserts into the top bosses, 3. caps into the top, 4. board onto the caps, 5. base over it, 6. four screws from below, hand-tight.

Firmware and flashing

Prompt for a coding agent (complete, no questions needed):

Write the firmware for a USB gamepad in Rust. Ask nothing; where a fact below is marked UNKNOWN, implement the stated safe behaviour and make the value a constant.

Hardware: Microchip ATmega32U4-AU (TQFP-44), 16 MHz crystal, 5 V from USB-C (VBUS through a 500 mA PTC), native USB 2.0 full speed. No other supply. Factory DFU bootloader in the boot section; HWB (PE2) is pulled to GND, so an external reset (reset button) starts the bootloader, power-on starts the application, provided the HWBE fuse is programmed as on parts shipped with the DFU bootloader (UNKNOWN for a given batch: if the reset button starts the application instead, report it).

Pins (MCU pin / port / function / notes):

- 8 PB0 UP, 9 PB1 DOWN, 10 PB2 LEFT, 11 PB3 RIGHT (D-pad); 30 PB6 A, 29 PB5 B, 28 PB4 X, 27 PD7 Y; 18 PD0 START, 19 PD1 SELECT. Buttons close to GND: inputs with internal pull-up, pressed = low.
- 31 PC6 LED P1, 32 PC7 LED P2, 26 PD6 LED ST: outputs, high = on, 3 mA each through 1 k.
- 3 D-, 4 D+ (22 Ohm series), 6 UCAP (1 uF), 7 VBUS: USB, handled by the USB controller.
- All unused pins (PE6, PB7, PD2, PD3, PD4, PD5, PF0, PF1, PF4 ... PF7, AREF): inputs with pull-up.
- PB1 ... PB3 double as the ISP lines (SCK, MOSI, MISO); never drive them as outputs.

Behaviour:

- Start: set the clock prescaler to 1 (CLKPR) regardless of the CKDIV8 fuse, start the USB PLL for the 16 MHz crystal, enable the watchdog (250 ms).
- USB HID gamepad: VID/PID constants (use the pid.codes test ID 1209:0001 until a real one is assigned), product string "USB gamepad", report every 1 ms (interval 1): 8 buttons A, B, X, Y, START, SELECT plus a hat switch from the D-pad (opposite directions pressed together count as neither). Report descriptor as a standard gamepad that Windows, Linux and macOS accept without a driver.
- Debounce: a change counts after 5 ms stable.
- LEDs: P1 on while enumerated and configured; P2 free for the host (output report, optional); ST blinks 2 Hz while not configured, off when configured.
- USB current: declare 50 mA in the configuration descriptor (real draw is below 30 mA).

Software: Rust nightly with the avr-atmega32u4 target (`-Zbuild-std=core`), avr-device and avr-hal (atmega-hal), USB through the usb-device crate with atmega-usbd as the bus driver and usbd-hid for the report. Pin the crate versions. Output an Intel HEX for the DFU bootloader. If the Rust USB stack does not enumerate reliably, report it with the log of what was tried instead of switching to another language.

Acceptance tests: host-side unit tests for debounce and the D-pad to hat conversion; a written manual test: the device enumerates as a gamepad on Windows and Linux (`evtest` / joy.cpl), every button maps to the right control, P1 lights when configured, unplug/replug ten times without a hang.

Flashing

Hardware: PC (Windows, macOS or Linux), USB-C cable, a paper clip for the reset hole in the bottom of the case.

Software: rustup with the nightly toolchain and rust-src, avr-gcc (linker), dfu-programmer. Windows: install the

libusb driver for the "ATm32U4DFU" device once with Zadig.

1. Build: cargo build --release, then avr-objcopy -O ihex target/avr-atmega32u4/release/gamepad.elf gamepad.hex.
2. Plug in USB-C, press the reset button through the hole in the bottom: the bootloader enumerates as ATm32U4DFU.
3. dfu-programmer atmega32u4 erase, then dfu-programmer atmega32u4 flash gamepad.hex.
4. dfu-programmer atmega32u4 launch (or unplug and plug in): the gamepad enumerates, LED P1 lights.
5. Check in the system's game-controller dialog (Windows joy.cpl, Linux evtest) that every button reacts.
6. Recovery: the bootloader cannot be erased over DFU. If it ever is, program it with an ISP programmer: SCK, MOSI, MISO on the DOWN, LEFT and RIGHT button pads, RESET on test point TP6, 5 V and GND on TP2 and TP3.

Instructions

Found impossible

- Nothing.

Open items (need the physical world)

- First prototype: check that every cap moves freely and the case closes flush.
- Firmware: confirm that the reset button starts the DFU bootloader (HWBE fuse).

Where to order

- Parts: the two distributor tables (fictional), one order each.
- PCB: Fab 1, settings below. Assembly: by hand; see the offers page for the make-or-buy comparison.

PCB order settings

- 2 layers, FR-4, 1.6 mm, 35 µm copper, ENIG, green solder mask, white silkscreen; min. track 0.2 mm, clearance 0.15 mm (USB-C pins only), min. drill 0.45 mm; quantity 25.
- Upload the Gerber/drill zip or the .kicad_pcb.

Deviations from the brief

- HS-2: TQFP-44 with 0.8 mm pitch and a USB-C receptacle with 0.85 mm pin pitch; no hand-friendlier MCU with native USB exists in a larger package.
- HS-4: clearance 0.2 mm (0.15 mm inside the USB-C footprint) instead of 0.5 mm; the TQFP pads force it.
- LB-1/LB-2: the board has no room next to the part for the value of C2, D1, D2, D3, F1, R3, R4, R7, U1; it is in the assembly drawing. R2, R9: reference only in the assembly drawing.
- JF-6: no spare solder fields; series product, the bottom is a ground pour.